MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
Regular EXAMINATION
October 2026
Fifth SEMESTER
Discipline Specific Elective (DSE) - MG5DSEBOTA08 - FORENSIC BOTANY
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. What is forensic palynology?

Forensic palynology is a branch of forensic science that deals with the study of pollen grains and spores to assist in criminal investigations. It uses the characteristics, distribution, and preservation of pollen and spores to establish links between a suspect, a victim, an object, and a crime scene. application of palynology (the scientific study of pollen and spores) to forensic investigations, particularly for determining the possible origin of materials and establishing associations between people, objects, and locations.

2. Name two sources from which forensic pollen samples may be collected

- Soil and sediment collected from a crime scene.
- Clothing and footwear of suspects or victims

3. What is a narcotic drug?

A narcotic drug is a substance that relieves pain and may cause drowsiness or stupor. Many narcotic drugs act on the central nervous system and can produce dependence with repeated use.

Examples: Opium, morphine, codeine, and heroin.

4. What is mid-range photography

It is a forensic photography technique used to photograph evidence at a moderate distance, showing both the evidence and its relationship to nearby objects or the crime scene.

Example: Photographing a knife along with the surrounding area where it was found.

5. What is the difference between centric and pennate diatoms?

Both types of diatoms are important in forensic limnology. Their identification and comparison with diatoms from water samples may help investigate suspected drowning cases and assess possible links between a body and a water source. However, diatom evidence must be interpreted alongside other findings. Centric diatoms usually circular, triangular, or polygonal in shape with Radial symmetry and commonly found in marine and freshwater plankton. Pennate diatoms usually elongated, boat-shaped, or needle-shaped with bilateral symmetry and commonly found attached to surfaces or in sediments; some are planktonic.

6. Mention two early uses of botanical evidence in forensic investigations

- Identification of the geographical origin of a suspect or object using plant materials such as pollen, seeds, and leaves.
- Linking suspects to crime scenes by comparing plant fragments or other botanical evidence found on clothing, footwear, or objects with plants at the scene.

7. Which plant is the source of ergot alkaloids

Ergot alkaloids are produced by the fungus *Claviceps purpurea*, which commonly infects rye (Secale cereale) and other grasses.

8. What is an introduced plant species?

An introduced plant species is a plant species that has been brought by humans, intentionally or accidentally, to an area outside its natural geographical distribution.

Example: *Lantana camara*, introduced into India from tropical America.

9. How can botanical evidence help in linking a victim to a crime scene

Botanical evidence helps link a victim to a crime scene by comparing plant materials such as pollen, spores, leaves, seeds, and wood fragments found on the victim's clothing, footwear, or body with plant materials collected from the crime scene. Similarities in these materials may indicate that the victim had contact with that location.

10. Mention two merits of RFLP

RFLP provides highly discriminating DNA profiles useful for individual identification. It can produce reliable results from good-quality DNA samples and is useful in forensic identification and paternity testing.

11. What is quality control?

Quality control is a procedure used to identify errors, defects, and variations in a product or process to ensure accuracy, reliability, and consistency. In a laboratory, quality control involves checking instruments, chemicals, reagents, and testing procedures to ensure that analytical results are accurate and reliable.

12. What is GC-MS?

GC-MS (Gas Chromatography–Mass Spectrometry) is an analytical technique used to separate, identify, and measure the chemical components present in a sample. It is used for detection and identification of drugs, poisons, and other chemical substances, identification of plant-derived toxins and chemical compounds in botanical samples and detection of toxic substances in blood, urine, and other biological samples.

[2x10 = 20]

Part B

Short Essays
Answer any **8** questions
Each question carries **5** marks

13. Explain the forensic significance of saliva analysis

Saliva analysis is the examination of saliva found at a crime scene to obtain biological evidence that may help identify individuals and establish links between suspects, victims, and crime scenes.

Forensic significance

1. DNA profiling: Saliva contains epithelial cells from the mouth that provide DNA for identifying individuals and comparing biological samples.
2. Identification of suspects: Saliva recovered from items such as cigarette butts, drinking glasses, bottles, and chewing gum may help associate a person with an object.
3. Sexual assault investigations: Saliva may be detected on the skin, clothing, or other items and can provide evidence of contact.
4. Detection of bite marks: Saliva deposited on skin or clothing during biting may provide DNA evidence linking a suspect to a victim.
5. Detection of biological stains: Saliva stains can be examined using presumptive tests, including tests for salivary amylase, followed by confirmatory analysis where available.
6. Drug and alcohol analysis: Saliva can be used to detect certain drugs and alcohol, depending on the substance and the analytical method employed.
7. Linking individuals to crime scenes: DNA obtained from saliva can help establish an association between a person and a crime-scene item, although it does not by itself prove when or how the saliva was deposited.

Common methods of saliva analysis

- Amylase tests: Used to screen for the presence of saliva.
- DNA profiling: Used for individual identification and comparison.
- Chemical and immunological methods: Used in certain cases to detect salivary components.
- Toxicological analysis: Used to detect drugs or alcohol in oral fluid.

14. Discuss the factors to be considered while transporting botanical evidences

Botanical evidence includes leaves, flowers, seeds, pollen, spores, wood, plant fragments, and other plant materials collected from a crime scene. Proper transportation is essential to preserve the evidence and ensure reliable forensic examination.

Important factors

- Proper packaging: Each botanical sample should be packed separately in a suitable container to prevent contamination and mixing of evidence. Paper bags or envelopes are generally suitable for dry plant materials.
- Prevention of contamination: Samples must be protected from contact with other plant materials, soil, chemicals, and biological evidence. Clean tools and gloves should be used during handling.
- Moisture control: Fresh plant materials may develop fungal growth or decay if sealed while wet. They should be transported using suitable ventilated packaging or preserved under appropriate conditions. Dry samples should be protected from moisture.
- Temperature control: Some samples, particularly fresh plant tissues intended for DNA analysis, may require cooling or freezing according to laboratory protocols. The appropriate temperature depends on the type of evidence and the intended analysis.
- Proper labelling: Every package should include the sample identification number, collection location, date and time of collection, name or identification of the collector, and other relevant details.
- Documentation: A complete record of the evidence collected, its condition, packaging, and transfer must be maintained. Photographs and collection notes may also be included.
- Chain of custody: Every transfer of evidence must be documented, including the identity of the person handing over and receiving the sample, date, time, and signatures where required.
- Prevention of physical damage: Fragile flowers, leaves, pollen-bearing structures, seeds, and other delicate materials must be packed carefully to prevent crushing, breakage, or loss.
- Protection from sunlight and heat: Direct sunlight and excessive heat may damage plant pigments, DNA, and other chemical components. Suitable protective packaging should be used.
- Transportation time: Samples should reach the forensic laboratory as quickly as practicable to minimize degradation, particularly when fresh plant tissues or unstable chemical compounds are involved.
- Special requirements for different samples: Pollen and spores may require airtight, clean containers to prevent loss and contamination, while bulky plant materials may require rigid packaging. Packaging must be appropriate to the specific sample and examination.
- Tamper-evident sealing: Packages should be securely sealed to prevent unauthorized access, accidental loss, or substitution during transportation.

15. How are flowers and seeds collected for forensic examination

Flowers and seeds are important forms of botanical evidence in forensic investigations. They may help link a suspect, victim, object, or vehicle to a particular crime scene. Flowers and seeds can help in plant species identification, linking a suspect or victim to a location, reconstructing

movements, and identifying potentially poisonous or controlled plant species. Their characteristics, including morphology, colour, size, and DNA profiles, may provide useful evidence.

Collection of flowers

- Photograph the evidence: Photograph flowers at the crime scene before collecting them, noting their position and surroundings.
- Use clean tools: Collect flowers using clean forceps, scissors, or other suitable tools to avoid contamination.
- Collect carefully: Collect whole flowers or relevant parts, including petals, sepals, stamens, pistils, and attached plant material, as required.
- Collect comparison samples: When appropriate, collect reference flowers from nearby plants for comparison.
- Package separately: Place delicate flowers in suitable rigid containers or paper packaging to prevent crushing. Fresh samples may require special preservation depending on the intended examination.
- Label properly: Record the sample number, collection location, date, time, collector's name, and relevant observations.

Collection of seeds

- Photograph the seeds: Document their location, distribution, and association with other evidence before collection.
- Collect using suitable tools: Use clean forceps, brushes, or scoops to collect seeds without damaging them.
- Collect individual and bulk samples: Collect seeds individually when their location is important, or collect a representative sample when numerous seeds are present.
- Collect reference samples: If necessary, collect seeds from identified plants in the surrounding area for comparison.
- Package appropriately: Place dry seeds in clean, dry paper envelopes or suitable rigid containers. Protect small seeds from loss during transport.
- Avoid mixing: Package samples from different locations separately to preserve their evidential value.
- Label and document: Record the exact collection site, date, time, sample description, and collector's details.

16. Explain the basic principles of forensic botany

Forensic botany is the branch of forensic science that applies botanical knowledge to criminal investigations. It involves the collection, identification, and analysis of plant materials to establish links between suspects, victims, objects, and crime scenes.

Basic principles of forensic botany

Locard's Exchange Principle-This principle states that whenever two objects come into contact, there is a possibility of transfer of materials between them. For example, pollen, seeds, leaves, or plant fragments may be transferred from a crime scene to a suspect's clothing or footwear.

Principle of Transfer of Botanical Evidence-Plant materials can be transferred through direct contact, wind, water, animals, or human activities. Such transfers may provide evidence of contact between a person or object and a particular environment.

Principle of Identification-Plant evidence can be identified by examining morphological, anatomical, microscopic, chemical, and molecular characteristics. These methods help determine the plant species or group to which the evidence belongs.

Principle of Comparison-Botanical samples collected from a crime scene can be compared with reference samples obtained from suspects, victims, vehicles, or other locations. Similarities and differences may help establish a possible association.

Principle of Individuality and Variation-Plants show variation in their morphological and genetic characteristics. Some botanical materials may possess distinctive features that help distinguish samples. However, similarities between plant samples do not necessarily prove that they originated from the same individual plant.

Principle of Distribution and Habitat-Different plant species have particular geographical distributions and ecological requirements. The presence of a plant species or a combination of species may help investigators infer the possible location or habitat associated with a crime.

Principle of Persistence and Degradation-Botanical evidence may remain on clothing, footwear, vehicles, or soil for varying periods. Its persistence depends on the type of material, environmental conditions, handling, and storage. Proper collection and preservation are essential to prevent degradation.

Principle of Scientific Examination and Interpretation-Botanical evidence must be examined using validated methods, appropriate controls, and reliable reference materials. Findings should be interpreted objectively, considering alternative explanations and the limitations of the evidence.

Principle of Proper Collection and Preservation-Botanical evidence must be collected, labelled, documented, packaged, and transported carefully. Separate packaging and appropriate storage conditions help prevent contamination, loss, and deterioration.

Principle of Chain of Custody-A continuous record must be maintained from the collection of botanical evidence to its presentation in court. This record documents who collected, handled, transferred, stored, and examined the evidence.

17. Explain two real-world applications of forensic palynology

Forensic palynology is the application of pollen and spore analysis to criminal investigations. Pollen grains and spores can help associate people, objects, or materials with particular locations because their distribution varies with vegetation, season, and habitat.

Linking a suspect to a crime scene

Pollen grains and spores may become attached to a suspect's clothing, footwear, hair, or vehicle. Forensic palynologists compare these samples with pollen collected from the suspected crime scene. Example: If pollen recovered from a suspect's footwear closely matches the pollen assemblage found in soil at a crime scene, it may support an association between the suspect and that location.

Tracing the origin of objects or materials

Pollen analysis can help determine the possible geographical or environmental origin of objects such as soil, illegally transported plant materials, or concealed items. Example: Pollen recovered from a package containing suspected illegally transported plant material may be compared with pollen assemblages from different geographical regions. The results may help narrow down the possible area of origin.

18. Define wildlife forensics and explain its scope

Wildlife forensics is a branch of forensic science that applies scientific techniques to investigate crimes involving wild animals, plants, and their products. It helps identify species, determine the origin of biological materials, and provide scientific evidence in cases of poaching, illegal wildlife trade, and violations of wildlife protection laws.

Scope of Wildlife Forensics

- Species identification: Identification of animal and plant species using morphological, anatomical, microscopic, and molecular techniques such as DNA analysis.
- Investigation of poaching: Examination of biological evidence, including hair, feathers, bones, blood, tissues, and other remains, to investigate illegal hunting and killing of wildlife.
- Illegal wildlife trade: Identification of products such as ivory, skins, horns, claws, shells, medicinal plant materials, and other illegally traded wildlife products.
- DNA analysis: Use of DNA profiling and DNA barcoding to identify species and, where possible, individual animals or populations.
- Plant-related wildlife crimes: Identification of protected and endangered plant species involved in illegal collection, transportation, and trade.
- Determination of geographical origin: Genetic and other scientific analyses may help determine the likely geographical origin of wildlife specimens and identify possible links to particular populations.

- Wildlife poisoning and mortality investigations: Analysis of tissues, stomach contents, and environmental samples to investigate suspected poisoning or unexplained wildlife deaths.
- Conservation of endangered species: Scientific evidence helps enforce conservation laws and protect threatened and endangered species from illegal exploitation.
- Habitat and ecological investigations: Examination of biological and environmental evidence can help investigate wildlife-related offences, habitat destruction, and other activities affecting protected ecosystems.
- Legal proceedings: Wildlife forensic experts prepare scientific reports and may present expert testimony in court to support investigations and prosecutions.

19. Discuss the documentation and chain-of-custody responsibilities of a forensic botanist.

A forensic botanist examines plant materials such as leaves, flowers, seeds, pollen, spores, wood, and other botanical evidence to assist in criminal investigations. Proper documentation and maintenance of the chain of custody are essential to ensure the integrity, reliability, and admissibility of botanical evidence in court.

Documentation responsibilities of a forensic botanist

Documentation is the systematic recording of information related to the collection, handling, examination, and analysis of botanical evidence.

Major responsibilities include:

- Recording crime-scene details: Document the date, time, location, environmental conditions, and circumstances of evidence collection.
- Photographic documentation: Photograph botanical evidence at the crime scene before collection, including close-up images and images showing its position in relation to surrounding objects.
- Sample identification: Assign a unique identification number to each sample and record its type, physical characteristics, quantity, and collection location.
- Collection records: Document the methods and tools used to collect samples, the name of the collector, and any precautions taken to prevent contamination.
- Packaging and labelling: Record the packaging materials, seal numbers, labels, and storage conditions used for each sample.
- Laboratory examination records: Maintain detailed notes on the methods, instruments, observations, results, reference samples, and quality-control procedures used during analysis.
- Preparation of forensic reports: Prepare clear, accurate, objective reports explaining the methods, findings, interpretations, and limitations of the examination.
- Maintaining confidentiality: Protect case records and sensitive investigative information from unauthorized access or disclosure.

Chain-of-custody responsibilities

The chain of custody is the continuous, chronological record of the collection, possession, transfer, storage, and examination of evidence from the crime scene until its presentation in court or final authorized disposition.

The forensic botanist has the following responsibilities:

- Proper collection: Collect botanical evidence using appropriate procedures to minimize contamination, loss, or damage.
- Unique labelling: Ensure that each sample is correctly identified and associated with the relevant case number.
- Secure packaging and sealing: Use suitable packaging and tamper-evident seals to protect the evidence and reveal unauthorized access.
- Recording every transfer: Document the names or identifiers of the persons transferring and receiving the evidence, along with the date, time, purpose, and required signatures.
- Secure storage: Store samples under appropriate conditions to prevent deterioration, contamination, or unauthorized access.
- Maintaining sample integrity: Avoid mixing samples and document any opening, repackaging, resealing, or change in storage conditions.
- Laboratory access control: Ensure that only authorized personnel handle the evidence and that all relevant access and handling are recorded according to laboratory procedures.
- Reporting discrepancies: Immediately document and report broken seals, missing samples, incorrect labels, or unexplained gaps in custody.
- Court testimony: Be prepared to explain the collection procedures, laboratory methods, custody records, and measures taken to preserve the evidence.

20. Discuss the limitations of botanical evidence in forensic investigations

Botanical evidence refers to plant materials such as leaves, flowers, seeds, fruits, pollen, spores, wood, and plant fragments used to investigate crimes. It can help link a suspect, victim, or object to a particular location. However, several limitations affect its reliability and interpretation in forensic investigations.

Widespread Distribution of Plant Species- Many plant species are widely distributed across different geographical regions. Finding a common plant species on a suspect or victim may not uniquely identify a crime scene.

Environmental Variability- Plant growth and distribution are influenced by soil type, temperature, rainfall, humidity, altitude, and seasonal changes. These factors can make comparisons between samples difficult.

Degradation of Evidence-Botanical evidence may deteriorate due to heat, moisture, sunlight, microbial activity, and decomposition. Damaged or decayed samples may be difficult to identify accurately.

Contamination- Plant materials may be transferred accidentally during transportation, handling, or contact with contaminated clothing and equipment. Contamination can produce misleading associations between individuals and crime scenes.

Difficulty in Species Identification- Closely related plant species may have similar morphological characteristics. Identification may require specialized expertise, microscopic examination, DNA analysis, or chemical tests.

Limited Reference Databases- Comprehensive reference collections and databases of plant species, pollen, spores, and plant DNA may not be available for every geographical region. This can restrict accurate identification and comparison.

Seasonal Variation-The availability of flowers, fruits, seeds, and pollen varies with the season. Therefore, botanical evidence collected at different times may not be directly comparable.

Difficulty in Establishing the Exact Time of Transfer- The presence of plant material on clothing or a vehicle does not necessarily indicate when the transfer occurred. Botanical evidence alone generally cannot establish the exact time of contact.

Problems in Establishing Geographic Origin- Although pollen, spores, and certain plant species can provide clues about geographical origin, similar species may occur in multiple locations. Consequently, botanical evidence may narrow down a location without identifying one exact place.

Complex Sample Collection and Preservation- Small samples such as pollen grains and plant fragments can be easily lost during collection. Improper packaging, storage, or transportation may damage evidence or introduce contamination. Limitations of Analytical Techniques- Microscopy, DNA analysis, chromatography, and spectroscopic methods may require expensive equipment, specialized training, and considerable time. Results may also be inconclusive when samples are degraded or present in very small quantities.

Difficulty in Interpreting Evidence- The presence of a plant fragment, pollen grain, or seed does not automatically prove that a person committed a crime. The material may have been acquired through innocent activities, secondary transfer, or ordinary environmental exposure.

Legal and Evidentiary Challenges- Botanical evidence must be collected, documented, preserved, and analysed using appropriate procedures. Poor chain-of-custody records, inadequate validation of analytical methods, or insufficient expert interpretation may reduce its evidentiary value in court.

Need for Corroborative Evidence- Botanical evidence is often most useful when combined with fingerprints, DNA profiling, soil analysis, witness statements, and other forensic findings. It may not independently establish guilt or prove the sequence of events.

21. Explain the importance of accurate documentation and reporting

Accurate documentation and reporting are essential in forensic science, including forensic botany. Documentation involves recording all relevant details of evidence collection, handling, preservation, and analysis. Reporting involves presenting the methods, observations, results, and conclusions in a clear, objective, and scientific manner.

Importance

- Maintains the integrity of evidence: Accurate records help ensure that botanical evidence, such as leaves, seeds, pollen, spores, and wood fragments, remains reliable throughout the investigation.
- Establishes chain of custody: Documentation records who collected, handled, transferred, stored, and analysed the evidence. This helps demonstrate that the evidence has not been improperly altered or substituted.
- Ensures traceability: Each sample should have a unique identification number, collection date, location, collector's name, and other relevant details. This allows investigators to trace the evidence throughout the forensic process.
- Supports accurate analysis: Detailed records of laboratory procedures, instruments, observations, and analytical results help other experts understand and verify the findings.
- Prevents errors and confusion: Proper labelling and systematic record-keeping reduce the risk of sample mix-ups, loss of evidence, and incorrect interpretation.
- Helps reconstruct crime scenes: Information about the location, distribution, and condition of botanical materials can help investigators establish possible links between suspects, victims, objects, and crime scenes.
- Promotes objectivity and transparency: Reports should clearly distinguish observations from interpretations and conclusions. Limitations and uncertainties must also be stated to avoid misleading claims.
- Supports legal proceedings: Well-prepared forensic reports provide scientific information that can be presented in court. Accurate documentation enables lawyers, judges, and other experts to evaluate the reliability of the evidence.
- Facilitates independent verification: Other forensic experts can review the records and, where possible, repeat the analytical procedures to assess the validity of the findings.
- Ensures professional accountability: Complete and accurate records demonstrate that forensic personnel followed established protocols, professional standards, and ethical guidelines.

22. Explain the role of plant metabolomics in forensic investigations

Plant metabolomics is the systematic study of small molecules called metabolites produced by plants. These include alkaloids, flavonoids, terpenoids, phenolic compounds, pigments, sugars, and organic acids. In forensic science, plant metabolomics helps identify plant materials, detect plant-derived toxins, compare botanical samples, and establish possible links between suspects, victims, and crime scenes.

Role of Plant Metabolomics in Forensic Investigations

1. Identification of plant species
   - Different plant species produce characteristic combinations of metabolites.
   - These chemical profiles can help distinguish closely related species that are difficult to identify using morphological characteristics alone.
   - This is particularly useful when plant samples are fragmented, powdered, or partially decomposed.
2. Detection of plant poisons

   - Metabolomic analysis helps identify toxic compounds present in poisonous plants.
   - It can support investigations involving plant poisoning, suspicious deaths, and the administration of plant-derived toxins.
   - Examples include alkaloids from Atropa belladonna, cardiac glycosides from Nerium oleander, and toxic compounds from Abrus precatorius.

3. Linking suspects to crime scenes

   - Plants growing in different locations may exhibit distinct chemical profiles because of differences in soil, climate, and environmental conditions.
   - Comparing the metabolite profiles of plant fragments recovered from a suspect's clothing with samples from a crime scene may provide evidence of a possible association.
   - However, chemical similarity alone cannot conclusively prove that a sample originated from one particular location.

4. Detection of drugs of plant origin

Plant metabolomics can help detect and characterize plant-derived drugs and their chemical constituents. Examples include:

   - Cannabis sativa — cannabinoids.
   - Opium poppy (Papaver somniferum) — morphine and codeine.
   - Tobacco (Nicotiana tabacum) — nicotine.
   - Coca plant (Erythroxylum coca) — cocaine.

5. Wildlife and illegal plant-trade investigations
   - Metabolite profiles can help distinguish certain plant species or plant products involved in illegal trade.
   - Chemical analysis may complement DNA barcoding and morphological identification.
   - This is useful in investigations involving protected medicinal plants, rare species, and adulterated botanical products.
6. Determination of plant origin and environmental history

- Metabolite composition can reflect environmental conditions, geographical origin, plant maturity, and physiological stress.
- Comparison of chemical profiles may help narrow down the possible origin of a botanical sample.
- Such conclusions require validated reference data because metabolite profiles can vary considerably within the same species.

7. Detection of adulteration and substitution
   - Metabolomic profiling can reveal unexpected chemical constituents in herbal products.
   - It can help identify substitution of one plant species for another and detect some forms of adulteration.
   - This is relevant to fraud investigations involving medicinal plants and commercial botanical products.

[5x8 = 40]

Part C

Essays

Answer any **1** questions

Each question carries **10** marks

23. Describe the common poisonous plants and their alkaloids

1. **Abrus precatorius**

Abrus precatorius belongs to the family Fabaceae (Leguminosae). It is commonly known as rosary pea, jequirity bean, or crab's eye. It is a climbing plant that produces attractive red seeds with a characteristic black spot. The seeds are highly poisonous and are of considerable importance in forensic toxicology. The principal toxic substance is abrin, a potent plant-derived toxin.

Toxic constituents

- Abrin: The principal toxic protein present in the seeds. It inhibits protein synthesis in cells, causing cellular damage and death.
- Abrine: An alkaloid found in the plant; it is not the principal cause of the severe toxicity associated with the seeds.
- Other constituents: The seeds contain additional compounds, but abrin is primarily responsible for their dangerous toxic effects.

2. **Cannabis sativa**

Cannabis sativa is a plant belonging to the family Cannabaceae. It is commonly known as cannabis, marijuana, or hemp. Certain varieties contain psychoactive substances that affect the central nervous system. The major psychoactive constituent is delta-9-tetrahydrocannabinol

(THC). Other important cannabinoids include cannabidiol (CBD), which is not intoxicating in the same way as THC.

Toxic constituents

- Delta-9-tetrahydrocannabinol (THC): The principal psychoactive compound responsible for intoxication and altered perception.
- Cannabidiol (CBD): A non-intoxicating cannabinoid that can interact with other medicines and cause adverse effects.
- Cannabinol (CBN): A cannabinoid formed partly through the degradation of THC.
- Other cannabinoids: May contribute to the overall pharmacological effects of cannabis.

3. **Claviceps purpurea**

Claviceps purpurea is a parasitic fungus belonging to the family Clavicipitaceae. It commonly infects rye and other grasses and cereals, replacing the grains with dark, hardened fungal structures called sclerotia. These sclerotia contain toxic alkaloids that cause a disease known as ergotism in humans and animals. Ergotism is an important topic in forensic toxicology because contaminated cereal products can cause severe poisoning.

Toxic constituents

- Ergotamine: Causes constriction of blood vessels and may reduce blood supply to tissues.
- Ergometrine (ergonovine): Produces contraction of uterine smooth muscle.
- Ergocristine: An ergot alkaloid that contributes to the pharmacological and toxic effects of ergot.
- Ergocryptine and ergocornine: Other alkaloids associated with ergot toxicity.

4. **Croton tiglium**

Croton tiglium belongs to the family Euphorbiaceae. It is commonly known as the purging croton. It is a medicinally known but highly toxic plant, and its seeds and seed oil are particularly dangerous. The plant is important in forensic toxicology because accidental or deliberate exposure may cause severe gastrointestinal irritation, dehydration and damage to tissues.

Toxic constituents

- Phorbol esters: Potent irritant compounds that activate protein kinase C and can cause inflammation and tissue damage.
- Croton oil: A strongly irritant oil obtained from the seeds that contains toxic phorbol esters.
- Crotonic acid and other constituents: Various compounds may contribute to the plant's irritant properties, but phorbol esters are particularly important in its toxicity.

5. **Atropa belladonna**

Atropa belladonna belongs to the family *Solanaceae* and is commonly known as deadly nightshade. It is a highly poisonous perennial herb containing tropane alkaloids that affect the central and peripheral nervous systems. All parts of the plant may be toxic, particularly the leaves, roots and berries. It is important in forensic botany because accidental ingestion or deliberate poisoning can produce severe anticholinergic toxicity.

Toxic constituents

- Atropine: Blocks muscarinic acetylcholine receptors, producing anticholinergic effects.
- Hyoscyamine: A potent antimuscarinic alkaloid that affects the nervous system and smooth muscles.
- Scopolamine (hyoscine): Produces anticholinergic effects and may cause marked sedation, confusion and memory disturbances.

6. **Gloriosa superba**

Gloriosa superba belongs to the family Colchicaceae. It is commonly known as the glory lily, flame lily, or climbing lily. It is a climbing perennial herb with striking red and yellow flowers and underground tubers. The plant is highly poisonous, particularly its tubers and seeds. It is important in forensic botany and toxicology because accidental or deliberate ingestion can cause severe and potentially fatal poisoning.

Toxic constituents

- Colchicine: The principal toxic alkaloid responsible for severe systemic toxicity.
- Colchicoside: A related alkaloid glycoside found in the plant.
- Other colchicine-related compounds: May contribute to the overall chemical properties of the plant.

7. **Jatropha curcas**

It is belongs to the family Euphorbiaceae. It is commonly known as physic nut, purging nut, or Barbados nut. It is a shrub widely cultivated in tropical and subtropical regions, particularly for its oil-rich seeds. The seeds are poisonous and may cause serious illness when ingested. The plant is important in forensic botany because accidental consumption, especially by children, can result in poisoning.

Toxic constituents

- Phorbol esters: Irritant compounds that alter cellular signalling and can produce inflammation and gastrointestinal toxicity.
- Curcin: A toxic protein found in the seeds that inhibits protein synthesis.
- Saponins: Compounds that may contribute to irritation and other biological effects.
- Other seed constituents: Various compounds may contribute to the overall toxicity of the plant.

8. **Nerium oleander**

Nerium oleander belongs to the family Apocynaceae. It is commonly known as oleander or rosebay. It is an evergreen ornamental shrub widely cultivated in gardens and along roadsides for its attractive flowers. All parts of the plant are poisonous, including the leaves, flowers, stems and sap. The plant contains cardiac glycosides that can cause serious and potentially fatal poisoning in humans and animals.

Toxic constituents

- Oleandrin: One of the major toxic cardiac glycosides responsible for oleander poisoning.
- Neriine: A cardiac glycoside associated with the toxicity of the plant.
- Other cardiac glycosides: Additional compounds contribute to its toxic effects.

9. **Nicotiana tabacum**

Nicotiana tabacum belongs to the family Solanaceae and is commonly known as the tobacco plant. It is cultivated widely for its leaves, which are used in the manufacture of cigarettes, cigars, chewing tobacco and other tobacco products. The plant contains nicotine, a potent toxic alkaloid that acts on the nervous system. Tobacco exposure can cause acute poisoning, dependence and several serious long-term health effects.

Toxic constituents

- Nicotine: The principal addictive alkaloid. It acts on nicotinic acetylcholine receptors and affects the central and peripheral nervous systems.
- Nornicotine: A related tobacco alkaloid with biological activity.
- Anabasine: A minor tobacco alkaloid that can also affect nicotinic acetylcholine receptors.
- Tobacco-specific nitrosamines: Harmful compounds formed during tobacco processing and curing; some are potent carcinogens.
- Combustion products: Tobacco smoke contains carbon monoxide, fine particles and numerous toxic chemicals, including carcinogens.

10. **Semecarpus anacardium**

Semecarpus anacardium belongs to the family Anacardiaceae and is commonly known as the marking nut tree or Indian marking nut. It is native to parts of India and is known for its fruit, which has been used traditionally for marking cloth and in certain traditional medicinal preparations. The plant's fruit contains an irritant oil that can cause significant skin inflammation and other toxic effects. It is important in forensic botany because exposure may occur accidentally or through inappropriate use of plant material.

Toxic constituents

- Bhilawanols: Phenolic compounds, including alkyl catechol derivatives, that can cause skin irritation and allergic contact dermatitis.
- Anacardic-acid-related compounds: Phenolic lipids that may contribute to the irritant properties of the plant material.
- Other phenolic compounds: Additional constituents may contribute to its biological activity.

11. **Strychnos nux-vomica**

Strychnos nux-vomica, commonly known as the strychnine tree, is a poisonous plant of forensic importance. Its seeds contain the toxic alkaloids strychnine and brucine, which affect the central nervous system, particularly the mechanisms responsible for inhibiting muscle activity.

Major toxic constituents

- Strychnine: The principal toxic alkaloid responsible for severe muscle spasms and convulsions.
- Brucine: Another toxic alkaloid, generally less potent than strychnine.
- These compounds are present mainly in the seeds.

12. **Thevetia nerifolia**

Thevetia neriifolia, commonly known as yellow oleander of the family Apocynaceae, is a highly poisonous ornamental plant. It is also commonly referred to as *Cascabela thevetia*. All parts of the plant are considered toxic, particularly the seeds.

Major toxic constituents are Thevetin A, Thevetin B, Peruvoside, Neriifolin

24. Describe any five famous forensic botany case histories and specify the botanical evidence used and its significance in each case.

1. The Charles Lindbergh Baby Kidnapping Case (1932)

In 1932, Charles Lindbergh's young son was kidnapped from his home in New Jersey, USA. A wooden ladder was used by the kidnapper to enter the child's bedroom. Bruno Richard Hauptmann was arrested and prosecuted for the crime.

Botanical evidence used:

- Wood samples from the ladder were examined by Arthur Koehler, a wood identification expert.
- Microscopic examination identified different types of wood, including pine, Douglas fir and birch.
- Wood-grain patterns, annual growth rings, knots and tool marks were studied.
- The marks on the ladder were compared with those made by tools found in Hauptmann's possession.

2. Italian Murder Case – Identification of a Body Concealment Site (2011)

In October 2011, skeletal remains of a woman were discovered near a river in a pre-Alpine valley in northern Italy. The woman's body was partially covered with leaves and branches. Her husband became a suspect in the murder investigation.

Botanical evidence used:

- Leaves and branches collected from above the body were identified as *Prunus laurocerasus* (cherry laurel).
- The species was not part of the natural vegetation growing at the discovery site.
- Similar garden waste was found approximately 100 metres away.

3. Swiss Murder Case – Identification of a Wooden Murder Weapon (2010)

In June 2010, skeletal remains of a missing girl were discovered in a woodland area in Canton Ticino, Switzerland. A fracture was observed in the frontal region of the victim's skull. Small pieces of wood were found embedded in the fracture.

Botanical evidence used:

- Wood fragments were collected from the skull fracture.
- Microscopic examination identified the fragments as compatible with *Castanea sativa* (sweet chestnut).
- These fragments were compared with chestnut wood blocks found nearby.
- The anthropological examination indicated that the skull injury was caused by a blunt object.

4. Italian Case – Estimation of Time Since Body Deposition (2011)

In July 2011, skeletal human remains were discovered in an uncultivated area near a main road in Italy. Plant roots were found growing in close association with the skeletal remains. Investigators wanted to estimate how long the body had remained at the location.

Botanical evidence used:

- Roots of *Phytolacca americana* (American pokeweed) were found penetrating the victim's shoe and growing around the skeletal remains.
- The growth and development of the roots were examined.
- The surrounding vegetation, including *Ligustrum* and *Sambucus nigra*, was also studied.
- Plant growth patterns and the estimated age of vegetation were used to determine the period of body deposition.

5. Italian Murder Case – Identification of the Primary Crime Scene (2011)

In February 2011, the partially skeletonised body of a missing girl was discovered in a field in northern Italy. Investigators suspected that the victim might have been murdered elsewhere and subsequently transported to the field. Botanical examination was conducted to determine whether the field was the original crime scene.
Botanical evidence used:

- Different plant species growing in the field were collected and identified.
- The vegetation found in the victim's right hand included:
    - *Sorghum halepense*
    - *Epilobium hirsutum*
    - *Rubus* species.
- A leaf of *Solidago gigantea* was found beneath the victim's skull.
- The distribution of vegetation beneath the body was compared with that of the surrounding field. **[10x1 = 10]**